

1. Overall Description

1.1 Functional Requirements Design

1.1.1 User Management Requirements

Requirement ID	Requirement Name	Description	Validation Rules
FR-001	User Registration	Allows anonymous users to create new accounts	Email uniqueness, password strength
FR-002	User Authentication	Allows registered users to authenticate and access their dashboard	Valid credentials, account verification
FR-003	Password Reset	Allows users to reset their password	Email verification

1.1.2 Course Management Requirements

Requirement ID	Requirement Name	Description	Validation Rules
FR-004	Create Course	Allows teachers to create new courses	Course title uniqueness, valid course data
FR-005	Edit Course	Allows teachers to edit existing courses	Valid course data
FR-006	Delete Course	Allows teachers to delete existing courses	User confirmation
FR-007	Enroll Student	Allows teachers to enroll students in a course	Valid student ID
FR-008	Upload Course Materials	Allows teachers to upload files and other learning materials to a course	File type and size validation

1.1.3 Session Management Requirements

Requirement ID	Requirement Name	Description	Validation Rules
FR-009	Create Session	Allows teachers to create new sessions	Session title uniqueness, valid session data
FR-010	Edit Session	Allows teachers to edit existing sessions	Valid session data

FR-011	Delete Session	Allows teachers to delete existing sessions	User confirmation
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1.1.4 Forum Management Requirements

Requirement ID	Requirement Name	Description	Validation Rules
FR-012	Create Thread	Allows users to create new threads in a forum	Thread title uniqueness, valid thread data
FR-013	Post Reply	Allows users to post replies to threads	Valid reply data

1.1.5 Chat Management Requirements

Requirement ID	Requirement Name	Description	Validation Rules
FR-014	Send Message	Allows users to send messages in a chat room	Valid message data
FR-015	Receive Message	Allows users to receive messages in a chat room	N/A

1.1.6 File Management Requirements

Requirement ID	Requirement Name	Description	Validation Rules
FR-016	Upload File	Allows users to upload files	File type and size validation
FR-017	Download File	Allows users to download files	N/A
FR-018	Delete File	Allows users to delete files	User confirmation

1.2 Use Case Design

1.2.1 Login

UC ID and Name:	
UC ID: UC-COM-01	Use Case Name: Login
Primary Actor:	Anonymous User
Trigger:	An Anonymous User initiates the login process by navigating to the Landing Page, open login modal and entering their credentials.
Description:	This use case describes the process of an Anonymous User logging into the full-teaching system using valid credentials. The system verifies the credentials and grants access to the appropriate dashboard based on the user's role.
Preconditions:	- The system must be online and able to process authentication requests.\n- The

	Anonymous User must have valid login credentials (username/email and password).
Postconditions:	- **Success:** The user is authenticated, their session is initiated, and they are redirected to the Dashboard Page.\n- **Failure:** If authentication fails, the system logs the failed attempt, notifies the user, and the user remains on the Landing Page. Access is denied.
Expected Results:	**Upon successful login:** The user is authenticated, their session is initiated, and they are redirected to the Dashboard Page.\n **Upon failed login:** If authentication fails, the system logs the failed attempt, notifies the user (e.g., displays an error message), and the user remains on login modal. Access is denied.
Normal Flow:	1. Anonymous User navigates to Landing Page, open login modal.\n2. Anonymous User enters their username (or email) and password into the designated fields.\n3. Anonymous User clicks the 'Login' button.\n4. System validates the format of the entered credentials (e.g., non-empty).\n5. System securely verifies the submitted credentials against the 'users' table in the database.\n a. If credentials are valid:\n - System establishes an authenticated session for the user.\n - System logs the successful login attempt, including user ID and timestamp.\n - System redirects the Staff to Dashboard Page.
Alternative Flows:	**AF1: Invalid Credentials** \n- At step 5, if credentials do not match a registered and active user:\n a. System logs the failed login attempt (e.g., username, IP address, timestamp).\n b. System displays an error message on login modal (e.g., "Invalid username or password. Please try again.\").\n c. User remains on login modal (returns to step 2).\n **AF2: Account Locked (Optional - if implemented)** \n- If the system implements account locking after multiple failed attempts:\n - At step 5, if credentials are valid but the account is locked:\n a. System displays an error message (e.g., "Your account is

	temporarily locked. Please try again later or contact support.\").\n - If credentials are invalid and this attempt triggers an account lock:\n a. System logs the failed attempt and locks the account.\n b. System displays an error message (e.g., \"Invalid credentials. Your account has been temporarily locked due to multiple failed attempts.\").	
Exceptions:	**E1: System Unavailable**\n- If the authentication service or database is unavailable at step 5:\n a. System logs the critical error.\n b. System displays a generic error message (e.g., \"Login service is temporarily unavailable. Please try again later.\").	
ID	Business Rule	Business Rule Description
BR-LOGIN-01	Valid Credentials Required	Users must enter valid login credentials (username/email and password) to access the system
BR-LOGIN-02	Role-Based Access Control	Upon successful login, the system grants access to the user's role-specific dashboard, ensuring that users can only access features and data authorized for their role.

1.2.2 User Registration

UC ID and Name:	
UC ID: UC-COM-02	Use Case Name: User Registration
Primary Actor:	Anonymous User
Trigger:	An Anonymous User initiates the registration process by open registration modal in navbar.
Description:	This use case describes the process of an Anonymous User registering for a new account in the full-teaching system.
Preconditions:	- The system must be online and able to process registration requests.
Postconditions:	- **Success:** A new user account is created, and the user is redirected to dashboard.\n- **Failure:** If registration fails, the user remains on registration modal.
Expected Results:	**Upon successful registration:** A new user account is created, and the user is

	redirected to dashboard.\n**Upon failed registration:** The user is notified of the error and remains on registration modal.
Normal Flow:	1. Anonymous User open registration modal in navbar.\n2. Anonymous User enters their desired username, email, password, and confirms the password in the designated fields.\n3. Anonymous User completes the reCAPTCHA validation.\n4. Anonymous User clicks the 'Register' button.\n5. System validates the format of the entered credentials (e.g., non-empty, valid email format, password strength, matching passwords).\n6. System securely creates a new user account in the 'users' table in the database.\n7. System redirects the user to dashboard.
Alternative Flows:	**AF1: Invalid Data**\n- At step 5, if the entered data is invalid:\n a. System displays an error message on registration modal (e.g., \"Invalid email format.\").\n b. User remains on registration modal (returns to step 2).\n**AF2: User Already Exists**\n- At step 6, if a user with the same username or email already exists:\n a. System displays an error message on registration modal (e.g., \"Username or email already exists.\").\n b. User remains on registration modal (returns to step 2).\n**AF3: Passwords Do Not Match**\n- At step 5, if the entered passwords do not match:\n a. System displays an error message on registration modal (e.g., \"Your passwords don't match!\").\n b. User remains on registration modal (returns to step 2).\n**AF4: reCAPTCHA Not Validated**\n- At step 4, if the reCAPTCHA is not validated:\n a. System displays an error message on registration modal (e.g., \"You must validate the captcha!\").\n b. User remains on registration modal (returns to step 2).
Exceptions:	**E1: System Unavailable**\n- If the registration service or database is unavailable at step 5:\n a. System logs the critical error.\n b. System displays a generic error message (e.g., \"Registration service is temporarily unavailable. Please try again later.\").

ID	Business Rule	Business Rule Description
BR-REG-01	Unique User Credentials	Username and emails must be unique.
BR-REG-02	Password Strength	Passwords must be between 8 and 20 characters long, and contain at least one digit, one lowercase letter, and one uppercase letter.
BR-REG-03	Password Confirmation	The entered password and confirmation password must match.
BR-REG-04	reCAPTCHA Validation	The user must complete the reCAPTCHA validation to prove they are not a robot.

1.2.3 Create Course

UC ID and Name:	
UC ID: UC-COM-03	Use Case Name: Create Course
Primary Actor:	Teacher
Trigger:	A Teacher initiates the course creation process by clicking the 'New Course' button.
Description:	This use case describes the process of a Teacher creating a new course in the full-teaching system.
Preconditions:	- The user must be logged in as a Teacher.
Postconditions:	- Success: A new course is created, and the Teacher is redirected to the course dashboard.\n- Failure: If course creation fails, the Teacher remains on the new course page.
Expected Results:	Upon successful creation: A new course is created, and the Teacher is redirected to the course dashboard.\n Upon failed creation: The Teacher is notified of the error and remains on the new course page.
Normal Flow:	1. Teacher clicks the 'New Course' button.\n2. System displays the new course page.\n3. Teacher enters the course title, image URL, and course information.\n4. Teacher clicks the 'Create' button.\n5. System validates the entered data (e.g., non-empty title).\n6. System creates a new course in the database.\n7. System redirects the Teacher to the course dashboard.
Alternative Flows:	AF1: Invalid Data \n- At step 5, if the entered data is invalid:\n a. System displays an error message on the new

	course page (e.g., \"Title cannot be empty.\").\n b. Teacher remains on the new course page (returns to step 3).	
Exceptions:	**E1: System Unavailable**\n- If the course creation service or database is unavailable at step 6:\n a. System logs the critical error.\n b. System displays a generic error message (e.g., \"Course creation service is temporarily unavailable. Please try again later.\").	
ID	Business Rule	Business Rule Description
BR-COURSE-01	Course Title Required	The course title cannot be empty.
BR-COURSE-02	Course Image Required	The course image URL cannot be empty.

1.2.4 Edit Course Information

UC ID and Name:	
UC ID: UC-COM-04	Use Case Name: Edit Course Information
Primary Actor:	Teacher
Trigger:	A Teacher initiates the course information editing process by clicking the 'Edit' button on the course information tab.
Description:	This use case describes the process of a Teacher editing the information of a course in the full-teaching system.
Preconditions:	- The user must be logged in as a Teacher.\n- The user must be on the course details page.
Postconditions:	- **Success:** The course information is updated.\n- **Failure:** If course information editing fails, the changes are not saved.
Expected Results:	**Upon successful editing:** The course information is updated.\n**Upon failed editing:** The Teacher is notified of the error and the changes are not saved.
Normal Flow:	1. Teacher clicks the 'Edit' button on the course information tab.\n2. System displays the course information in an editable format.\n3. Teacher modifies the course information.\n4. Teacher clicks the 'Save' button.\n5. System validates the entered data.\n6. System updates the course information in the database.\n7. System displays the updated course information.
Alternative Flows:	**AF1: Invalid Data**\n- At step 5, if the entered data is invalid:\n a. System

	displays an error message.\n b. Teacher remains in editing mode (returns to step 3).	
Exceptions:	**E1: System Unavailable** \n- If the course editing service or database is unavailable at step 6:\n a. System logs the critical error.\n b. System displays a generic error message (e.g., \"Course editing service is temporarily unavailable. Please try again later.\").	
ID	Business Rule	Business Rule Description
BR-COURSE-03	Course Information Validation	The course information cannot be empty.

1.2.5 Manage Students

UC ID and Name:	
UC ID: UC-COM-05	Use Case Name: Manage Students
Primary Actor:	Teacher
Trigger:	A Teacher initiates the student management process by navigating to the 'Students' tab in the course details view.
Description:	This use case describes the process of a Teacher managing the students enrolled in a course. This includes adding new students (one by one, in bulk, or via file import) and removing existing students from the course.
Preconditions:	- The user must be logged in as a Teacher.\n- The user must be on the course details page.
Postconditions:	- **Success:** The student list for the course is updated.\n- **Failure:** If student management fails, the student list remains unchanged.
Expected Results:	**Upon successful management:** The student list for the course is updated.\n **Upon failed management:** The Teacher is notified of the error and the student list remains unchanged.
Normal Flow:	**NF-A: Add Single Student:** \n **The Teacher selects "Add Single Student". \n **The system displays a form to enter a student's email. \n **The Teacher enters the student's email address. \n **The Teacher clicks the 'Add' button. \n **The system validates the entered email address. \n **The system adds the student to the course. \n **The system displays the updated student list.

	NF-B: Add Multiple Students:\nThe Teacher selects 'Add Multiple Students'.\n**The system displays a textarea for entering a list of emails.\n**The Teacher pastes a list of email addresses (separated by comma, semicolon, or new line).\n**The Teacher clicks the 'Add Students' button.\n**The system processes the entire list of emails.\n**The system adds all valid students who are not already enrolled.\n**The system displays a summary report (e.g., "Successfully added 8/10 students. 2 students failed.") and refreshes the student list. NF-C: Import from File:\n**The Teacher selects "Import from File".\n**The system displays an interface for file upload.\n**The Teacher selects and uploads a file (e.g., .csv, .txt) containing a list of emails, one per line. \n**The Teacher clicks the 'Import' button. \n**The system reads and processes the file. \n**The system adds all valid students from the file who are not already enrolled. \n**The system displays a summary report (similar to NF-B) and refreshes the student list. NF-D: Removing a student: \n**The Teacher clicks the 'Remove' button next to a student's name. \n**The system prompts the Teacher for confirmation. \n**The Teacher confirms the removal. \n**The system removes the student from the course. \n**The system displays the updated student list.
Alternative Flows:	**AF1: Invalid Email**\n- At step 5, if the entered email address is invalid:\n a. System displays an error message.\n b. Teacher remains on the add student form (returns to step 3).\n**AF2: Student Not Found**\n- At step 6, if the student with the entered email address is not found:\n a. System displays an error message.\n b. Teacher remains on the add student form (returns to step 3).\n**AF3: Student Already Enrolled**\n- At step 6, if the student is already enrolled in the course:\n a. System displays an error message.\n b. Teacher remains on the add student form (returns to step 3).

Exceptions:	**E1: System Unavailable** \n- If the student management service or database is unavailable:\n a. System logs the critical error.\n b. System displays a generic error message (e.g., \"Student management service is temporarily unavailable. Please try again later.\").	
ID	Business Rule	Business Rule Description
BR-STUDENT-01	Valid Email Required	The email address of the student must be valid.
BR-STUDENT-02	Student Must Exist	The student must have a registered account in the system.
BR-STUDENT-03	Student Not Already Enrolled	The student must not already be enrolled in the course.

1.2.6 Manage Sessions

UC ID and Name:	
UC ID: UC-COM-06	Use Case Name: Manage Sessions
Primary Actor:	Teacher
Trigger:	A Teacher initiates the session management process by navigating to the 'Sessions' tab in the course details view.
Description:	This use case describes the process of a Teacher managing the sessions for a course. This includes adding new sessions, editing existing sessions, and deleting sessions.
Preconditions:	- The user must be logged in as a Teacher.\n- The user must be on the course details page.
Postconditions:	- **Success:** The session list for the course is updated.\n- **Failure:** If session management fails, the session list remains unchanged.
Expected Results:	**Upon successful management:** The session list for the course is updated.\n **Upon failed management:** The Teacher is notified of the error and the session list remains unchanged.
Normal Flow:	**Adding a session:** \n1. Teacher clicks the 'New Session' button.\n2. System displays a form for adding a new session.\n3. Teacher enters the session title, description, date, and time.\n4. Teacher clicks the 'Add' button.\n5. System validates the entered data.\n6. System adds the session to the course.\n7. System displays the updated session

	list.\n\n**Editing a session:**\n1. Teacher clicks the 'Edit' button next to a session.\n2. System displays a form with the existing session data.\n3. Teacher modifies the session data.\n4. Teacher clicks the 'Save' button.\n5. System validates the entered data.\n6. System updates the session.\n7. System displays the updated session list.\n\n**Deleting a session:**\n1. Teacher clicks the 'Delete' button next to a session.\n2. System prompts the Teacher for confirmation.\n3. Teacher confirms the deletion.\n4. System deletes the session.\n5. System displays the updated session list.	
Alternative Flows:	**AF1: Invalid Data**\n- If the entered data is invalid:\n a. System displays an error message.\n b. Teacher remains on the form.	
Exceptions:	**E1: System Unavailable**\n- If the session management service or database is unavailable:\n a. System logs the critical error.\n b. System displays a generic error message (e.g., \"Session management service is temporarily unavailable. Please try again later.\").	
ID	Business Rule	Business Rule Description
BR-SESSION-01	Session Title Required	The session title cannot be empty.
BR-SESSION-02	Session Date Required	The session date cannot be empty.
BR-SESSION-03	Session Time Required	The session time cannot be empty.

1.2.7 Manage Forum

UC ID and Name:	
UC ID: UC-COM-07	Use Case Name: Manage Forum
Primary Actor:	Teacher, Student
Trigger:	A user initiates the forum management process by navigating to the 'Forum' tab in the course details view.
Description:	This use case describes the process of a user managing the forum for a course. This includes adding new entries and replying to existing entries.
Preconditions:	- The user must be logged in.\n- The user must be enrolled in the course.\n- The forum must be activated by the teacher.

Postconditions:	- **Success:** The forum is updated.\n- **Failure:** If forum management fails, the forum remains unchanged.	
Expected Results:	**Upon successful management:** The forum is updated.\n **Upon failed management:** The user is notified of the error and the forum remains unchanged.	
Normal Flow:	**Adding an entry:** \n1. User clicks the 'New Entry' button.\n2. System displays a form for adding a new entry.\n3. User enters the entry title and content.\n4. User clicks the 'Add' button.\n5. System validates the entered data.\n6. System adds the entry to the forum.\n7. System displays the updated forum.\n\n **Replying to an entry:** \n1. User clicks the 'Reply' button on an entry.\n2. System displays a form for replying to the entry.\n3. User enters the reply content.\n4. User clicks the 'Reply' button.\n5. System validates the entered data.\n6. System adds the reply to the entry.\n7. System displays the updated forum.	
Alternative Flows:	**AF1: Invalid Data** \n- If the entered data is invalid:\n a. System displays an error message.\n b. User remains on the form.	
Exceptions:	**E1: System Unavailable** \n- If the forum management service or database is unavailable:\n a. System logs the critical error.\n b. System displays a generic error message (e.g., \"Forum management service is temporarily unavailable. Please try again later.\").	
ID	Business Rule	Business Rule Description
BR-FORUM-01	Entry Title Required	The entry title cannot be empty.
BR-FORUM-02	Entry Content Required	The entry content cannot be empty.

1.2.8 Manage Files

UC ID and Name:	
UC ID: UC-COM-08	Use Case Name: Manage Files
Primary Actor:	Teacher
Trigger:	A Teacher initiates the file management process by navigating to the 'Files' tab in the course details view.
Description:	This use case describes the process of a Teacher managing the files for a course. This includes adding new files and file

	groups, editing existing files and file groups, and deleting files and file groups.
Preconditions:	- The user must be logged in as a Teacher.\n- The user must be on the course details page.
Postconditions:	- Success: The file structure for the course is updated.\n- Failure: If file management fails, the file structure remains unchanged.
Expected Results:	Upon successful management: The file structure for the course is updated.\n Upon failed management: The Teacher is notified of the error and the file structure remains unchanged.
Normal Flow:	Adding a file group: \n1. Teacher clicks the 'New Folder' button.\n2. System displays a form for adding a new file group.\n3. Teacher enters the file group title.\n4. Teacher clicks the 'Add' button.\n5. System validates the entered data.\n6. System adds the file group to the course.\n7. System displays the updated file structure.\n\n Adding a file: \n1. Teacher clicks the 'New File' button.\n2. System displays a form for adding a new file.\n3. Teacher selects the file to upload.\n4. Teacher clicks the 'Add' button.\n5. System validates the selected file.\n6. System uploads the file and adds it to the course.\n7. System displays the updated file structure.\n\n Editing a file or file group: \n1. Teacher clicks the 'Edit' button next to a file or file group.\n2. System displays a form with the existing data.\n3. Teacher modifies the data.\n4. Teacher clicks the 'Save' button.\n5. System validates the entered data.\n6. System updates the file or file group.\n7. System displays the updated file structure.\n\n Deleting a file or file group: \n1. Teacher clicks the 'Delete' button next to a file or file group.\n2. System prompts the Teacher for confirmation.\n3. Teacher confirms the deletion.\n4. System deletes the file or file group.\n5. System displays the updated file structure.
Alternative Flows:	AF1: Invalid Data \n- If the entered data is invalid:\n a. System displays an error

		message.\n b. Teacher remains on the form.
Exceptions:		**E1: System Unavailable**\n- If the file management service or database is unavailable:\n a. System logs the critical error.\n b. System displays a generic error message (e.g., \"File management service is temporarily unavailable. Please try again later.\").
ID	Business Rule	Business Rule Description
BR-FILE-01	File Group Title Required	The file group title cannot be empty.
BR-FILE-02	File Required	A file must be selected for upload.

1.2.9 Participate in a Video Session

UC ID and Name:	
UC ID: UC-COM-09	Use Case Name: Participate in a Video Session
Primary Actor:	Teacher, Student
Trigger:	A user joins a video session by clicking on a session link.
Description:	This use case describes how a user (either a teacher or a student) participates in a live video session. It covers video and audio streaming, chat functionality, and requests for intervention.
Preconditions:	- User must be logged in.\n- User must be enrolled in the course.\n- The session must be currently active.
Postconditions:	- User has participated in the video session and exits the session.
Expected Results:	The user is able to see and hear the teacher (and any other broadcasting student), send and receive chat messages, and request to speak. If a student, their request to speak can be granted or denied by the teacher.
Normal Flow:	**Common Actions:** 1. User joins the session and connects to the video and chat services. 2. User can send and receive messages in the chat. 3. User can view the list of connected participants. 4. User can control the video playback (toggle fullscreen, play/pause, adjust volume). 5. User clicks the 'Exit' button to leave the session.

	Student-Specific Actions: 1. Student clicks the 'Ask for Intervention' button to request permission to speak. 2. If the Teacher grants permission, the student's audio and video are broadcast to all participants. The main video view switches to the student. 3. The student can click the 'Cancel Intervention' button to stop broadcasting. **Teacher-Specific Actions:** 1. Teacher's audio and video are broadcast by default upon joining. 2. Teacher sees a list of students requesting intervention. 3. Teacher can grant or revoke a student's intervention request.		
Alternative Flows:	**AF1: Intervention Request Denied** - If the Teacher denies a student's intervention request, the student is notified but their audio/video are not broadcast. **AF2: Teacher Leaves Session** - If the Teacher leaves the session, all video streams are terminated.		
Exceptions:	**E1: Connection Lost** - If the user's connection to the video or chat service is lost, the system will attempt to reconnect. If reconnection fails, the user is notified and may need to rejoin the session.		
ID	Business Rule	Business Rule Description	
BR-VIDEO-01	Teacher-Led Session	Only the teacher's audio and video are broadcast by default. Students must request permission to speak.	
BR-VIDEO-02	Intervention Control	The teacher has the authority to grant or revoke a student's permission to speak at any time.	

1.2.10 View Calendar

UC ID and Name:	
UC ID: UC-COM-10	Use Case Name: View Calendar
Primary Actor:	Teacher, Student
Trigger:	A user navigates to the 'Calendar' page.

Description:	This use case describes how a user views their upcoming sessions on a calendar.	
Preconditions:	- User must be logged in.	
Postconditions:	- The user has viewed their session schedule.	
Expected Results:	The user's calendar is displayed, showing all their scheduled sessions. The user can switch between month, week, and day views, and can navigate to a session's course page.	
Normal Flow:	1. User navigates to the 'Calendar' page.\n2. The system retrieves all the sessions for the user's courses.\n3. The system displays the sessions on a calendar.\n4. The user can switch between month, week, and day views.\n5. The user can click on a session to view its details.\n6. The user can click a link to navigate to the session's course page.	
Alternative Flows:	**AF1: No Sessions** - If the user has no scheduled sessions, the calendar is displayed empty.	
Exceptions:	**E1: System Unavailable** - If the calendar service or database is unavailable:\n a. System logs the critical error.\n b. System displays a generic error message (e.g., \"Calendar service is temporarily unavailable. Please try again later.\").	
ID	Business Rule	Business Rule Description
BR-CALENDAR-01	Session Visibility	Users can only see the sessions for the courses they are enrolled in.

1.2.10 Logout

UC ID and Name:	
UC ID:	UC-COM-11
Use Case Name:	Logout
Primary Actor:	Logged-in User (Teacher, Student)
Trigger:	The user clicks the 'Logout' button in the navigation bar.
Description:	This use case describes the process of a logged-in user securely ending their session and logging out of the system.
Preconditions:	- The user must be logged into the system.
Postconditions:	- Success: The user's session is terminated. The user is redirected to the Landing Page. - Failure: The user remains logged in.

Expected Results:	- Upon successful logout: The system terminates the user's session and redirects them to the Landing Page. - Upon failed logout: The user is notified of an error and remains on the current page.
Normal Flow:	1. User clicks the 'Logout' button in the user menu. 2. System confirms the logout request. 3. System terminates the current user session, clearing any temporary authentication data. 4. System redirects the user to the Landing Page.
Alternative Flows:	None.

2. External Interface Requirements

2.1 User Interface Design

2.1.1 Detailed UI Elements by Screen

2.1.1.1 SC-001: Landing Page

Element	Type	Description
Navbar	Navbar	Reference to 2.1.2.1 for more informations
Application Title	Heading	The main title of the page: "FullTeaching".
Slogan	Heading	A secondary heading with the text: "Rediscover the joy of learning".
Welcome Button	Button	A large button with the text "Welcome!" that opens the login modal.
Image Slider	Slider	A slider component that cycles through three images, each with a caption.
Informational Text	Text/Paragraph	Various sections of text describing the application's features and benefits.
Parallax Backgrounds	Image	Multiple sections with dynamic background images that create a parallax scrolling effect.

2.1.1.1.1 Login Modal

Element	Type	Description
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Welcome Message	Heading	The main title of the page: "Welcome to FullTeaching !"
Email	Input Field	An input field for the user's email.
Password	Input Field	An input field for the user's password.
Log In Button	Button	Submits the login credentials and logs the user in.
Close Button	Button	Closes the login modal.
Sign Up Link	Link	A link with the text "SIGN UP" that navigates to the registration form for new users. It is preceded by the text "Not registered yet?".

2.1.1.1.2 Registration Modal

Element	Type	Description
Page Title	Heading	The main title of the modal: "Welcome to FullTeaching!"
Username	Input Field	An input field for username
Email	Input Field	An input field for the user's email address.
Name	Input Field	An input field for the user's name.
Password	Input Field	An input field for the user's password.
Confirm Password	Input Field	An input field for confirming the user's password.
reCAPTCHA	Checkbox	A reCAPTCHA checkbox to prevent spam.
Sign Up Button	Button	Submits the registration form. Navigates to the Dashboard Page after successful registration and login.
Log In Link	Link	A link to navigate to open login modal for existing users.

2.1.1.2 SC-004: Dashboard Page

Element	Type	Description
Navbar	Navbar	Reference to 2.1.2.1 for more informations
My Courses Title	Heading	A title for the user's course list.

Add New Course Button	Button	A circular button with a plus icon for teachers to open the "New Course" modal.
Course List	List/Link	A list of the user's courses. Each item displays the course image and title. Navigate to Course Details Page when clicked.
Edit Course Button	Button	An edit icon next to each course for teachers, which opens the "Modify Course" modal.
My Calendar Title	Heading	A title for the user's calendar.
Calendar	Calendar	A calendar displaying scheduled sessions.
Calendar Month/Year	Text	Displays the current month and year being viewed.
Calendar Navigation	Buttons	Buttons with arrows to navigate to the previous/next month and a button to return to the current date.
Session Indicator	Icon	A red circle on a day, indicating the number of sessions scheduled.
Session Link	Icon/Link	When a day is clicked, an arrow icon appears for each session, linking to the course's "Sessions" tab.

2.1.1.2.1 Add New Course Modal (from Dashboard)

Element	Type	Description
Modal Title	Heading	The title of the modal: "New course".
Course Name	Input Field	An input field for the new course's name.
Course Image	File Input	A file input for uploading an image for the new course.
Cancel Button	Button	A button to close the modal without creating a new course.
Send Button	Button	A button to submit the form and create the new course.

2.1.1.2.2 Modify Course Modal (from Dashboard)

Element	Type	Description
Modal Title	Heading	The title of the modal: "Modify course".

Course Name	Input Field	An input field for editing the course's name.
Course Image	File Input	A file input for changing the course's image.
Allow Deletion	Checkbox	A checkbox that must be ticked to enable the "Delete" button.
Delete Button	Button	A button to delete the course.
Cancel Button	Button	A button to close the modal without saving any changes.
Send Button	Button	A button to submit the form and save the changes to the course.

2.1.1.3 SC-005: Course Details Page

Element	Type	Description
Navbar	Navbar	Reference to 2.1.2.1 for more informations
Back Button	Button	A circular button with a left arrow that navigates the user back to the Dashboard.
Course Title	Heading	The title of the course.
Tabs	Tabs	A set of tabs for navigating course sections: Info, Sessions, Forum, Files, and Students.

2.1.1.5.1 Info Tab

Element	Type	Description
Course Information	Text Area	Displays the course information.
Edit Button	Button	A button for teachers to enter editing mode.
Rich Text Editor	Editor	(Editing Mode) The text area becomes a rich text editor for formatting.
Preview Button	Button	(Editing Mode) A button to preview the formatted text.
Save Button	Button	(Editing Mode) A button to save the changes.
Cancel Button	Button	(Editing Mode) A button to discard changes and exit editing mode.

2.1.1.5.2 Sessions Tab

Element	Type	Description
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Session List	List	A list of the course's sessions, showing title, description, and date.
New Session Button	Button	A button for teachers to open the "New Session" modal.
Edit Session Button	Button	A button next to each session for teachers to open the "Modify Session" modal.

2.1.1.5.2.1 New/Modify Session Modal (from Sessions Tab)

Element	Type	Description
Modal Title	Heading	The title of the modal (e.g., "New session" or "Modify session").
Session Title	Input Field	An input field for the session's title.
Session Description	Text Area	A text area for the session's description.
Date Picker	Date Input	A date picker to select the date of the session.
Time Picker	Time Input	A time picker to select the time of the session.
Delete Button	Button	(Edit mode only) A button to delete the session.
Cancel Button	Button	A button to close the modal without saving changes.
Send Button	Button	A button to submit the form and save the session.

2.1.1.5.3 Forum Tab

Element	Type	Description
Entry List	List	A list of the forum's entries, showing title, author, and date.
New Entry Button	Button	A button to open the "New Entry" modal.
Comment List	List	A list of comments for a selected entry.
Reply Button	Button	A button to reply to a specific comment.

2.1.1.6.3.1 New Entry Modal (from Forum Tab)

Element	Type	Description
Modal Title	Heading	The title of the modal: "New entry".
Entry Title	Input Field	An input field for the new entry's title.

Entry Comment	Text Area	A text area for the initial content of the entry.
Cancel Button	Button	A button to close the modal without creating the entry.
Send Button	Button	A button to submit the form and create the new entry.

2.1.1.5.4 Files Tab

Element	Type	Description
File Tree	Tree View	A tree view of the course's files and file groups.
New File Group Button	Button	A button for teachers to open the "New File Group" modal.
Add File Button	Button	A button for teachers to open the "New File" modal.
Edit/Delete Buttons	Buttons	Buttons for teachers to edit or delete file groups.
Add file button	Button	A button for teachers to open the "Add files" modal.
Add Subgroup	Button	A button for teachers to open the "New subgroup" modal.

2.1.1.5.4.1 New File Group Modal (from Files Tab)

Element	Type	Description
Modal Title	Heading	The title of the modal: "New file group".
File group title	Input Field	An input field for the new folder's name.
Cancel Button	Button	A button to close the modal.
Send Button	Button	A button to create the new file group.

2.1.1.5.4.2 Add files Modal (from Files Tab)

Element	Type	Description
Modal Title	Heading	The title of the modal: "Add files".
File Uploader	File Input	An area to drag and drop or select a file for upload.
Cancel Button	Button	A button to remove file uploaded.
Upload Button	Button	A button to upload the selected file.
Close Button	Button	A button to close "Add files" modal.

2.1.1.5.5 Students Tab

Element	Type	Description
Student List	List	A list of enrolled students, showing their name and picture.
Add Student Button	Button	A button for teachers to open the "Add new attenders" modal.
Delete Student Button	Button	A button next to each student for teachers to remove them from the course.

2.1.1.5.5.1 Add New Attenders Modal (from Students Tab)

Element	Type	Description
Modal Title	Heading	The title of the modal: "Add new attenders".
Tabs	Tabs	Tabs to switch between adding students: "Simple", "Multiple", and "From File".
Email Input	Input Field	(Simple Tab) An input field for a single student's email.
Emails Text Area	Text Area	(Multiple Tab) A text area for multiple student emails.
Separator Input	Input Field	(Multiple Tab) An input to specify a custom separator for the email list.
File Uploader	File Input	(From File Tab) An uploader for a text file of emails.
Cancel Button	Button	A button to close the modal.
Send Button	Button	A button to add the specified students.

2.1.1.4 SC-006: Video Session Screen

Element	Type	Description
Navbar	Navbar	Reference to 2.1.2.1 for more informations
Video Player	Video Player	Displays the main video stream (teacher or presenting student).
Small Video Player	Video Player	Displays a smaller, secondary video stream.
Video Controls	Buttons/Slider	Controls for play/pause, fullscreen, and volume.
Side Menu Button	Button	A button to open/close the side menu.

Side Menu	Side Navigation	A side menu containing the chat and the list of connected users.
Chat	Chat Interface	A chat for sending and receiving messages.
Connected Users List	List	A list of users currently connected to the session.
Ask for Intervention	Button	A button for students to request permission to speak.

2.1.2 Common UI Components

2.1.2.1 Navbar

2.1.2.1.1 Navbar before authenticated

Element	Type	Description
Sign Up button	button	Open Registration Modal
Login button	button	Open Login Modal

2.1.2.1.2 Navbar after authenticated

Element	Type	Description
Course button	button	Navigate to Dashboard Page
Dropdown list	Dropdown	Open dropdown list
Logout button	Dropdown item	Logout user from system

Appendix

BUSINESS FLOW 1 (SEQUENTIAL UPDATE): AN END-TO-END USER JOURNEY

This flow represents a sequential, step-by-step journey on the FullTeaching platform. It is designed to test a logical chain of actions, from course creation by a teacher to student interaction and completion of learning activities.

Phase 1: Teacher Initializes and Sets Up the Course

1. **Login and Course Creation:** A Teacher with a pre-provisioned account accesses the Landing Page and performs a **Login (UC-COM-01)**. From their Dashboard, they initiate the **Create Course (UC-COM-03)** process, entering a title and basic information.
2. **Populate Course Content:** After being redirected to the course details page, the Teacher performs the following tasks:
 - Navigates to the 'Info' tab to **Edit Course Information (UC-COM-04)**.

- Moves to the 'Sessions' tab to **Manage Sessions (UC-COM-06)**, creating a schedule for live video classes.
 - Accesses the 'Files' tab to **Manage Files (UC-COM-08)**, uploading necessary materials.
3. **Logout:** After completing the course setup, the Teacher performs a **Logout (UC-COM-11)**.
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Phase 2: Student Registers and is Enrolled

1. **Account Registration:** A new Student visits the platform and completes the **User Registration (UC-COM-02)** process. Her dashboard is empty as she is not yet enrolled in any courses. She then logs out.
 2. **Teacher Enrolls Student:** The Teacher logs back in (**Login (UC-COM-01)**). They navigate to the created course, go to the 'Students' tab, and use the **Manage Students (UC-COM-05)** use case to add the new Student using her email address.
 3. **Logout:** After successfully enrolling the student, the Teacher performs a **Logout (UC-COM-11)**.
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Phase 3: Student Interacts with the Course

1. **Course Exploration:** The Student performs a **Login (UC-COM-01)**. She now sees the new course on her Dashboard. She clicks to enter the details page, reviews the course information, schedule, and downloads materials.
 2. **Forum Interaction:** Having a question, the Student navigates to the 'Forum' tab and uses the **Manage Forum (UC-COM-07)** use case to post a new question.
 3. **Logout:** After posting her question and reviewing the course content, the Student performs a **Logout (UC-COM-11)**.
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Phase 4: Teacher Responds and Updates the Course

1. **Respond and Manage:** The Teacher performs a **Login (UC-COM-01)** and notices the new forum post.
 - They enter the 'Forum' to answer the student's question (**UC-COM-07**).
 - While logged in, they decide to update the description of an upcoming session (**UC-COM-06**).

- They also upload an additional reference document (**UC-COM-08**).
 - 2. **Logout:** After completing these management tasks, the Teacher performs a **Logout (UC-COM-11)**.
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Phase 5: Participating in a Live Class

1. **Teacher Starts Session:** When it's time for the class, the Teacher performs a **Login (UC-COM-01)**, navigates to the 'Sessions' tab, and clicks the link to start the video session, initiating **Participate in a Video Session (UC-COM-09)**.
 2. **Student Joins Session:** Shortly after, the Student also performs a **Login (UC-COM-01)** and joins the same video session (**UC-COM-09**).
 3. **In-Session Interaction:** The Student asks a question via chat and later requests to speak. The Teacher grants permission, allowing the Student's audio and video to be broadcast.
 4. **End Session:** When the class concludes, they both click the 'Exit' button. The Student performs a **Logout (UC-COM-11)** first.
 5. The Teacher also performs a **Logout (UC-COM-11)** after ensuring the session has ended.
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Phase 6: Student Reviews and Plans

1. **Use Calendar:** A day later, the Student performs a **Login (UC-COM-01)** to plan her week. She uses the **View Calendar (UC-COM-10)** feature to review all scheduled sessions and navigate to the course page.
2. **Final Logout:** After planning her schedule, the Student performs a final **Logout (UC-COM-11)**.

This action concludes the full, sequential interactive learning cycle on the FullTeaching platform.

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BUSINESS FLOW 2: USER ACCOUNT RECOVERY SCENARIO

This flow describes the sequential process when a registered user (Teacher or Student) forgets their password and needs to regain access to the FullTeaching platform. This is a critical support function, ensuring that users can securely recover their accounts. This use case corresponds to the **Password Reset (FR-003)** functional requirement.

1. **Initiate Request:** A user accesses the Landing Page and opens the Login Modal. Realizing they have forgotten their password, they click the "Forgot Password?" link.
2. **Email Verification:** The system displays a form requesting the user to enter their registered email address. The user enters their email and submits the request.
3. **Send Recovery Link:** The system validates that the email exists in the database. If it exists, the system sends an email containing a unique, time-sensitive password recovery link to that address. The user is notified on-screen to check their inbox.
4. **Access Reset Page:** The user opens the email and clicks the recovery link. This action opens a secure password reset page on the FullTeaching platform.
5. **Create New Password:** On this page, the user is prompted to enter a new password and confirm it. The system validates the new password against the defined strength rules (referencing **BR-REG-02: Password Strength**).
6. **Completion and Login:** After the user submits a valid new password, the system updates the user's credentials and displays a success message. The user is then redirected to the login page, where they can perform a **Login (UC-COM-01)** with their new password.

BUSINESS FLOW 3: COURSE LIFECYCLE MANAGEMENT - DELETING A COURSE

This flow outlines the process for a Teacher when they need to permanently delete an unused course from the platform. This is a critical administrative action that requires careful confirmation to prevent unintentional data loss. This use case corresponds to the **Delete Course (FR-006)** functional requirement.

1. **Login and Access Dashboard:** The Teacher performs a **Login (UC-COM-01)** to access the system. From their Dashboard, they can see the list of courses they manage.
2. **Open Course Edit Modal:** The Teacher identifies the course to be deleted and clicks the 'Edit Course' icon next to it. This action opens the "Modify Course" modal.
3. **Enable Deletion:** Inside the modal, there is an "Allow Deletion" checkbox. For safety, the "Delete" button is disabled until the Teacher ticks this checkbox.
4. **Final Confirmation:** After enabling the "Delete" button, the Teacher clicks it. The system displays a final confirmation dialog, warning that this action is irreversible and all course-related data (sessions, files, student lists, forum posts, etc.) will be permanently deleted.
5. **Execute Deletion:** The Teacher confirms the action in the dialog. The system proceeds to delete the course and all its associated data from the database.

6. **UI Update:** Upon successful deletion, the system automatically closes the modal and returns the Teacher to the Dashboard. The deleted course no longer appears in the list.
 7. **Logout:** After completing the administrative task, the Teacher performs a **Logout (UC-COM-11)**.
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BUSINESS FLOW 4: BULK STUDENT MANAGEMENT AND REMOVAL

- **Phase 1: Preparation**

1. **Teacher Login:** A Teacher logs into the system (**UC-COM-01**).
2. **Navigate to Course:** The Teacher navigates to a pre-existing course.
3. **Prepare Student Data:** Prepare a list of 4 student emails:
 - student_valid_1@example.com (registered user, not yet enrolled)
 - student_valid_2@example.com (registered user, not yet enrolled)
 - invalid-email (invalid email format)
 - student_already_enrolled@example.com (a student who is already in the course)
4. Prepare a .txt file containing the emails student_valid_3@example.com and student_not_registered@example.com.

- **Phase 2: Test Multi-Email Enrollment**

1. **Navigate to Students Tab:** The Teacher opens the 'Students' tab for the course.
2. **Initiate Bulk Add:** The Teacher selects the "Add Multiple Students" feature.
3. **Enter Data:** The Teacher pastes the prepared list of 4 emails into the textarea.
4. **Submit & Verify:** The Teacher clicks the 'Add' button.
 - **Expected Result:** The system should report 2 successful enrollments (student_valid_1, student_valid_2) and 2 failures (due to invalid format and already enrolled errors). The student list is updated correctly.

- **Phase 3: Test Import from File Enrollment**
 1. **Initiate File Import:** The Teacher selects the "Import from File" feature.
 2. **Upload File:** The Teacher uploads the prepared .txt file.
 3. **Submit & Verify:** The Teacher clicks the 'Import' button.
 - **Expected Result:** The system should report 1 successful enrollment (student_valid_3) and 1 failure (due to the student account not being registered). The list is updated again.
 - **Phase 4: Test Student Removal**
 1. **Select Student to Remove:** The Teacher finds student_valid_1@example.com in the list.
 2. **Initiate Removal:** The Teacher clicks the 'Remove' (or 'Delete') button next to the student's name.
 3. **Confirm Action:** The system displays a confirmation dialog. The Teacher clicks 'Confirm'.
 4. **Verify Removal:**
 - **Expected Result:** The student student_valid_1 disappears from the course's student list.
 - **Phase 5: Cleanup**
 1. **Logout:** The Teacher logs out of the system (**UC-COM-11**).
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BUSINESS FLOW 5: COURSE STRUCTURE MANAGEMENT (DELETION & ORGANIZATION)

- **Phase 1: Preparation**
 1. **Teacher Login:** A Teacher logs into the system (**UC-COM-01**).
 2. **Navigate to Course:** The Teacher navigates to a course that contains at least one session and one file.
- **Phase 2: Manage and Delete Files**
 1. **Navigate to Files Tab:** The Teacher opens the 'Files' tab.

2. **Create File Group:** The Teacher uses the "New Folder" feature to create a new file group named "Reference Materials" (**UC-COM-08**).
 - **Expected Result:** A new folder appears in the file tree.
 3. **Delete File:** The Teacher locates an existing file and clicks its 'Delete' button.
 4. **User Confirmation (File):** The system displays a confirmation dialog. The Teacher clicks 'Confirm'.
 - **Expected Result:** The file is deleted from the list. (BR-FR018-1 is tested).
- **Phase 3: Manage and Delete Sessions**
 1. **Navigate to Sessions Tab:** The Teacher opens the 'Sessions' tab.
 2. **Delete Session:** The Teacher locates an existing session and clicks its 'Delete' button.
 3. **User Confirmation (Session):** The system displays a confirmation dialog. The Teacher clicks 'Confirm'.
 - **Expected Result:** The session is deleted from the list. (BR-FR011-1 is tested).
 - **Phase 4: Cleanup**
 1. **Logout:** The Teacher logs out of the system (**UC-COM-11**).
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BUSINESS FLOW 6: COURSE LIFECYCLE (COURSE DELETION)

- **Phase 1: Preparation**
 1. **Teacher Login:** A Teacher logs into the system (**UC-COM-01**).
 2. **Create a Temporary Course:** The Teacher creates a new course specifically for this test, named "Temporary Course for Deletion" (**UC-COM-03**).
- **Phase 2: Test Course Deletion Process**
 1. **Navigate to Dashboard:** The Teacher returns to the main Dashboard that lists all courses.

2. **Initiate Edit/Delete:** The Teacher finds the "Temporary Course for Deletion" and clicks the 'Edit' button (or settings icon) to open the modify modal.
 3. **Attempt to Delete (without confirmation):** In the modal, the Teacher clicks the 'Delete' button **without** ticking the 'Allow Deletion' checkbox.
 - **Expected Result:** The 'Delete' button should be disabled or have no effect.
 4. **Enable Deletion:** The Teacher ticks the 'Allow Deletion' checkbox.
 5. **Initiate Deletion:** The Teacher clicks the 'Delete' button.
 6. **Final Confirmation:** The system displays a final confirmation dialog (e.g., "Are you sure you want to permanently delete this course?"). The Teacher clicks 'Confirm'.
 - **Expected Result:** The "Temporary Course for Deletion" disappears from the Dashboard. (BR-FR006-1 is tested).
- **Phase 3: Cleanup**
 1. **Logout:** The Teacher logs out of the system (**UC-COM-11**).